

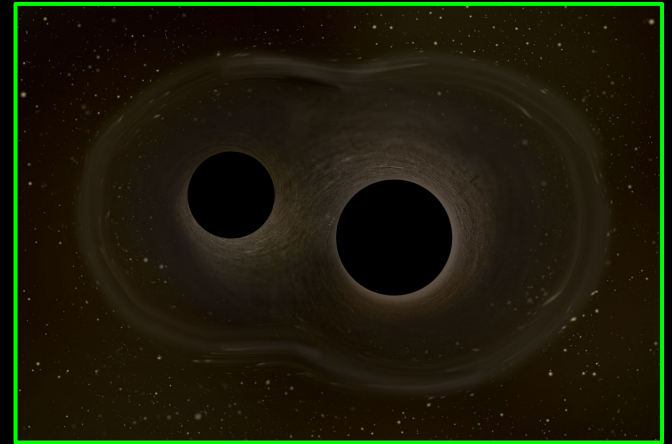
Correcting Compact Binary Inspiral Gravitational Waveform Uncertainty in Bayesian Inference Parameter Estimation

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Compact Binary Inspiral Gravitational Waves

- Predicted by General Relativity
- Produced by binary black holes, neutron stars, and white dwarves
- Measured by interferometers
 - LIGO Hanford
 - LIGO Livingston
 - Virgo
 - Kagra
- Signals received as gravitational wave strain, h

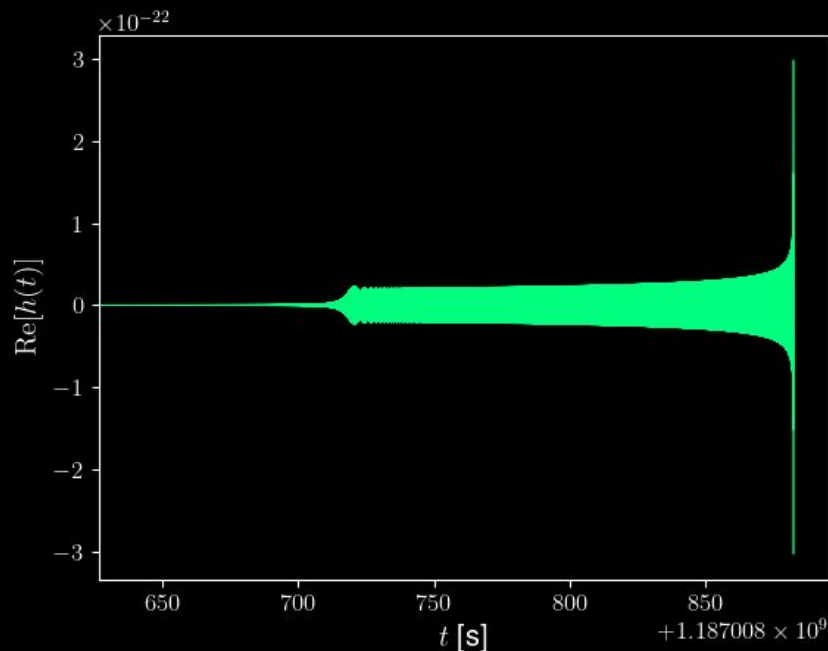
$$h(t) \sim \frac{\Delta L(t)}{L}$$



Time Domain Gravitational Waves

- Strain, h , as a function of time
- Quadrupolar Radiation
- Intuitive
- Define coalescence time and phase
- Define time domain amplitude, A , and phase, ψ

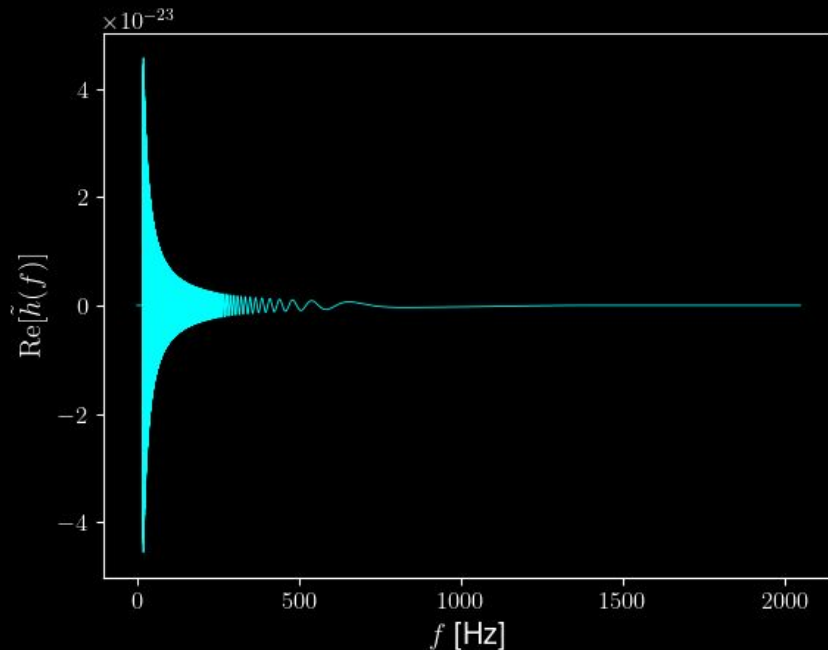
$$h(t) = A(t)e^{i\psi(t)}$$



Frequency Domain Gravitational Waves

- Strain, h , as a function of frequency
- Fourier transform of the time domain waveform
- Most often expressed as its amplitude and phase rather than the full waveform
- Used in parameter estimation
- Define frequency domain amplitude, $\mathcal{A}$, and phase, ϕ

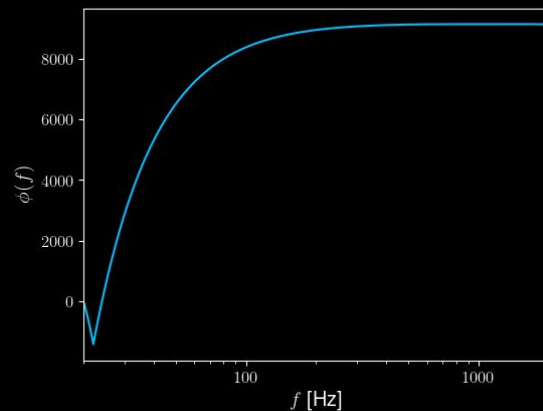
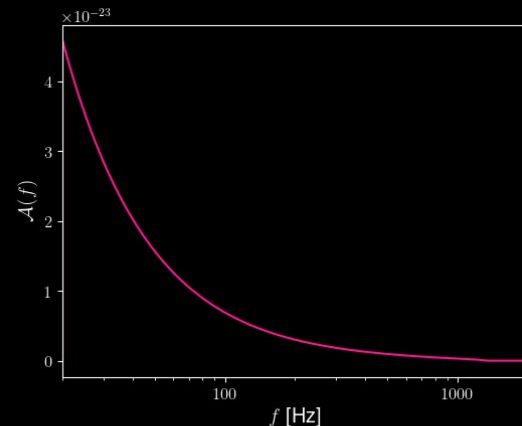
$$\tilde{h}(f) = \mathcal{A}(f)e^{i\phi(f)}$$



Frequency Domain Gravitational Waves

$$\mathcal{A}(f) = \sqrt{\tilde{h}^*(f)\tilde{h}(f)} = |\tilde{h}(f)|$$

$$\phi(f) = \tan^{-1} \left(\frac{\text{Im}[\tilde{h}(f)]}{\text{Re}[\tilde{h}(f)]} \right)$$



Frequency Domain Waveform Models

TaylorF2 Waveform Model:

$$\mu_{\text{TF2}}(f; \theta) = \mathcal{A}_{\text{TF2}}(f; \theta) e^{i\phi_{\text{TF2}}(f; \theta)}$$

$$\mathcal{A}_{\text{TF2}}(f; \theta) = \sqrt{\frac{2\eta}{3\pi^{\frac{1}{3}}}} f^{-\frac{7}{6}}$$

$$\phi_{\text{TF2}}(f; \theta) = 2\pi f t_c - \phi_c - \frac{\pi}{4} + \frac{3}{128\eta} (\pi M f)^{-\frac{5}{3}} \sum_{i=0}^7 \phi_i (\pi M f)^{\frac{i}{3}}$$

Basics of Parameter Estimation

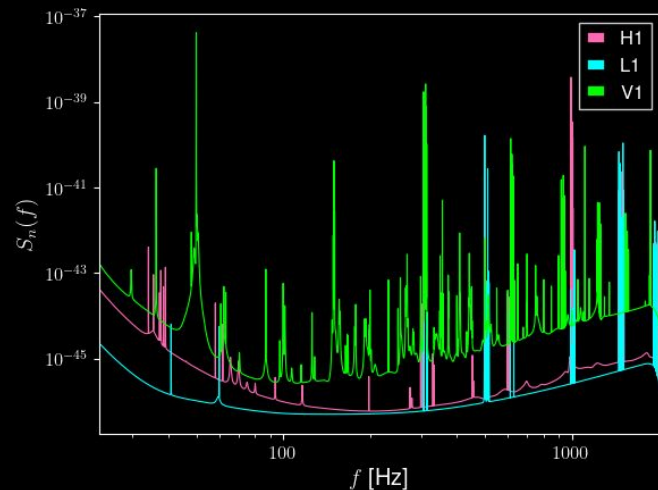
- Bayesian inference
- Given gravitational wave data, h
- Frequency domain waveform model, μ , which is parameterized by source parameters, θ
- Prior distributions are given for the source parameters
- Bayesian inference sampler, like [bilby](#), chooses random θ values according to the prior and generates a waveform to compare to h
- Likelihood function peaks when the data and model match
- Bayes' Theorem provides the posterior, or output, distributions of the source parameters

$$P(\theta|h) = \frac{\mathcal{L}(h|\theta)\pi(\theta)}{\mathcal{Z}(h)}$$

Basics of Parameter Estimation

$$\mathcal{L}(h|\theta) = \prod_j \frac{1}{2\pi S_n(f_j)} \exp \left(-2\Delta f \frac{|\tilde{h}(f_j) - \mu(f_j; \theta)|^2}{S_n(f_j)} \right)$$

Symbol	Description
μ	FD Waveform Model
S_n	Power Spectral Densities
θ	Source Parameters
Δf	Step Frequency, $f_j - f_{j-1}$



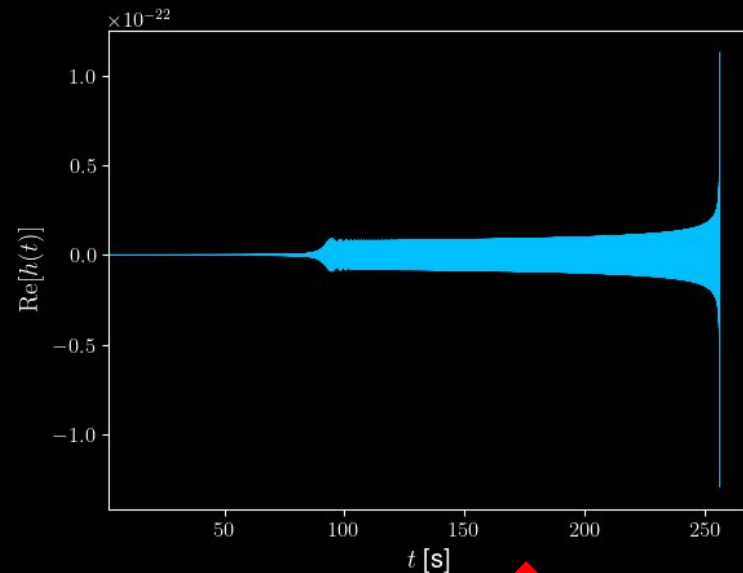
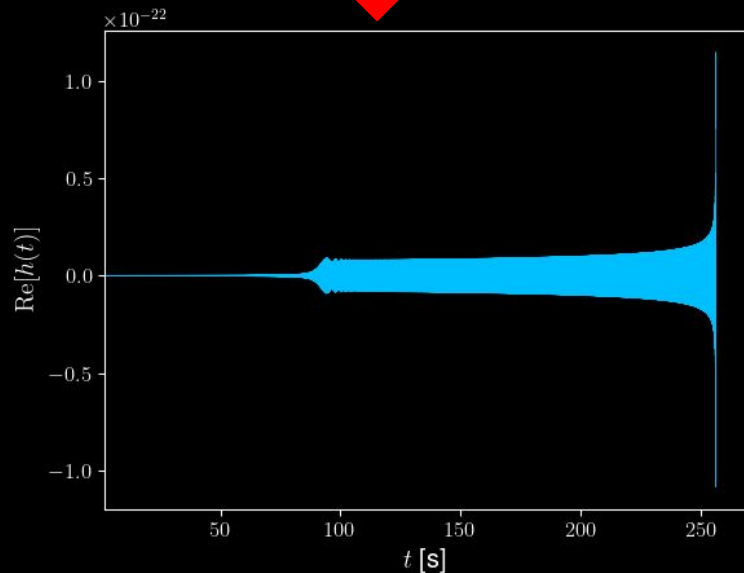
Waveform Uncertainty

- Errors between our models and general relativity
- Errors between general relativity and the true waveform
 - beyond-GR effects
- Posterior distributions thrown off by these errors

$$G_{\mu\nu} = 8\pi T_{\mu\nu}$$

Waveform Uncertainty

$$m_1 = 2.42, m_2 = 0.82, \chi_1 = -0.05, \chi_2 = 0.05, \Lambda_1 = 10, \Lambda_2 = 1000$$



$$m_1 = 1.45, m_2 = 1.31, \chi_1 = 0.500, \chi_2 = 0.00, \Lambda_1 = 600, \Lambda_2 = 0.01$$

Modifying the Waveform Model

1. Express the model in terms of its amplitude and phase

$$\mu(f; \theta) = \mathcal{A}(f; \theta) e^{i\phi(f; \theta)}$$

2. Add deviations to the amplitude and phase

$$\mu'(f; \theta) = (\mathcal{A}(f; \theta) + d\mathcal{A}(f)) e^{i(\phi(f; \theta) + d\phi(f))}$$

3. Define the deviations in terms of useful quantities

$$d\mathcal{A}(f) = ??? \quad d\phi(f) = ???$$

Frequency Domain Waveform Differences

$$\Delta\mathcal{A}(f) = \frac{\mathcal{A}_2(f) - \mathcal{A}_1(f)}{\mathcal{A}_1(f)}$$

$$\Delta\phi(f) = \phi_2(f) - \phi_1(f) - 2\pi f t_c - \phi_c$$

Modifying the Waveform Model

1. Express the model in terms of its amplitude and phase

$$\mu(f; \theta) = \mathcal{A}(f; \theta) e^{i\phi(f; \theta)}$$

2. Add deviations to the amplitude and phase

$$\mu'(f; \theta) = (\mathcal{A}(f; \theta) + d\mathcal{A}(f)) e^{i(\phi(f; \theta) + d\phi(f))}$$

3. Define the deviations in terms of useful quantities: the waveform differences

$$d\mathcal{A}(f) = \mathcal{A}(f; \theta) \Delta\mathcal{A}(f)$$

$$d\phi(f) = \Delta\phi(f)$$

Modifying the Waveform Model

4. Substitute in the new definitions of the deviations $\mu'(f; \theta) = (\mathcal{A}(f; \theta) + \mathcal{A}(f; \theta)\Delta\mathcal{A}(f))e^{i(\phi(f; \theta) + \Delta\phi(f))}$
5. Factor out the amplitude and distribute the i $\mu'(f; \theta) = \mathcal{A}(f; \theta)(1 + \Delta\mathcal{A}(f))e^{i\phi(f; \theta) + i\Delta\phi(f)}$
6. Split the exponential into two parts $\mu'(f; \theta) = \mathcal{A}(f; \theta)(1 + \Delta\mathcal{A}(f))e^{i\phi(f; \theta)}e^{i\Delta\phi(f)}$

Modifying the Waveform Model

7. Rearrange the amplitude and phase

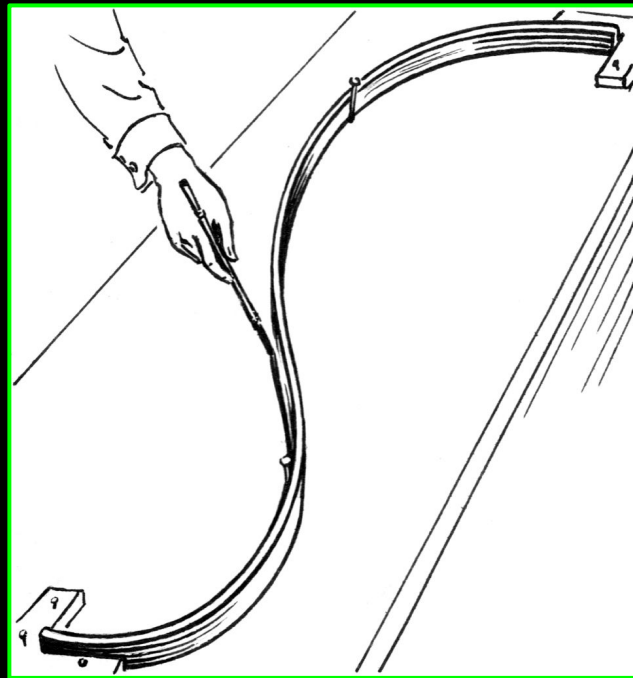
$$\mu'(f; \theta) = \mathcal{A}(f; \theta) e^{i\phi(f; \theta)} (1 + \Delta\mathcal{A}(f)) e^{i\Delta\phi(f)}$$

8. Substitute in $\mu(f; \theta) = \mathcal{A}(f; \theta) e^{i\phi(f; \theta)}$

$$\mu'(f; \theta) = \mu(f; \theta) (1 + \Delta\mathcal{A}(f)) \exp[i\Delta\phi(f)]$$

Parameterizing the Waveform Differences

- The waveform differences cannot be sampled directly
- To use the new waveform model with **bilby**, each waveform difference function must be described by a small number of parameters.
- We use **cubic splines** for C^1 continuity
 - Continuous and differentiable
 - Described by points, or nodes



Parameterizing the Waveform Differences

$$\Delta\mathcal{A}(f) \rightarrow \Delta\mathcal{A}_\delta(f; \{f_k, \alpha_k\})$$

Fixed Frequency Nodes

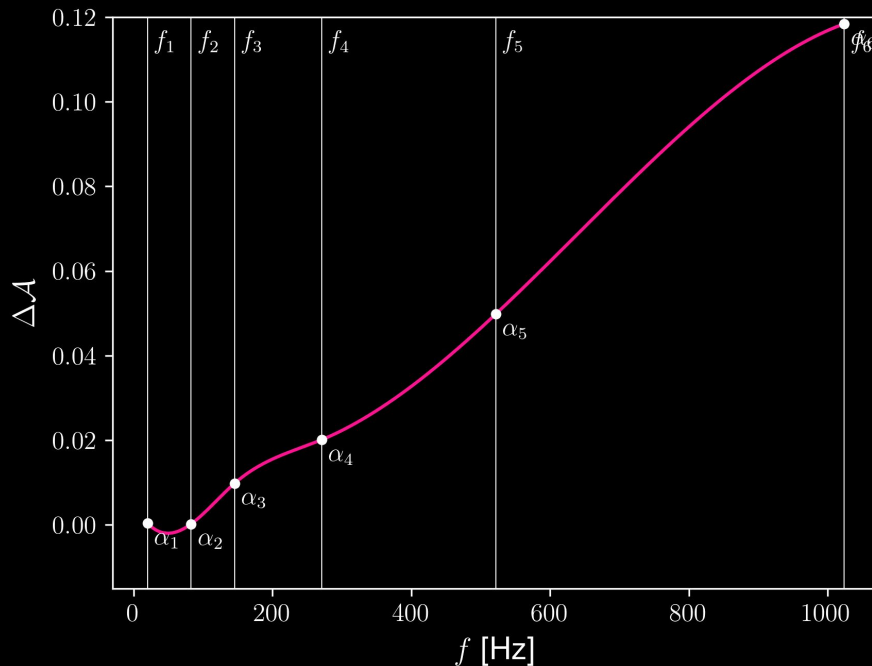
Spline Parameters

$$\Delta\phi(f) \rightarrow \Delta\phi_\delta(f; \{f_k, \varphi_k\})$$

Parameterizing the Waveform Differences

$$\Delta\mathcal{A}(f) \rightarrow \Delta\mathcal{A}_\delta(f; \{f_k, \alpha_k\})$$

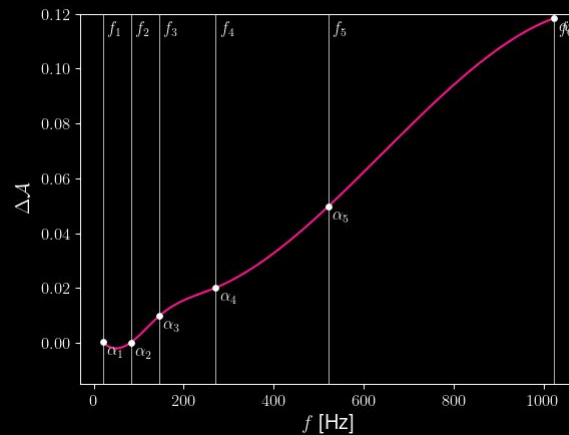
- Frequency nodes are fixed
- Spline parameters lie along the frequency nodes
- Cubic functions fill the gaps
- Whole function described by the spline parameters



Parameterizing the Waveform Differences

$$\Delta\mathcal{A}(f) \rightarrow \Delta\mathcal{A}_\delta(f; \{f_k, \alpha_k\})$$

$$\Delta\phi(f) \rightarrow \Delta\phi_\delta(f; \{f_k, \varphi_k\})$$



$$\mu_c(f; \theta, \alpha, \varphi) = \mu(f; \theta)(1 + \Delta\mathcal{A}_\delta(f; \{f_k, \alpha_k\})) \exp[i\Delta\phi_\delta(f; \{f_k, \varphi_k\})]$$

Likelihood Function

$$\mathcal{L}_{\mathcal{C}}(h|\theta, \alpha, \varphi) = \prod_j \frac{1}{2\pi S_n(f_j)} \exp \left(-2\Delta f \frac{|\tilde{h}(f_j) - \mu(f; \theta) \cdot \nu(f; \alpha, \varphi)|^2}{S_n(f_j)} \right)$$

$$\nu(f; \alpha, \varphi) = (1 + \Delta \mathcal{A}_{\delta}(f; \{f_k, \alpha_k\})) \exp[i\Delta \phi_{\delta}(f; \{f_k, \varphi_k\})]$$

Defining the Prior

- To perform parameter estimation, we need a prior for the waveform uncertainty parameters
- Plot many sets of waveform differences between two approximants
 - This will show the distribution of the differences across frequency
- From this distribution, we choose the form of the prior

Waveform Model Differences

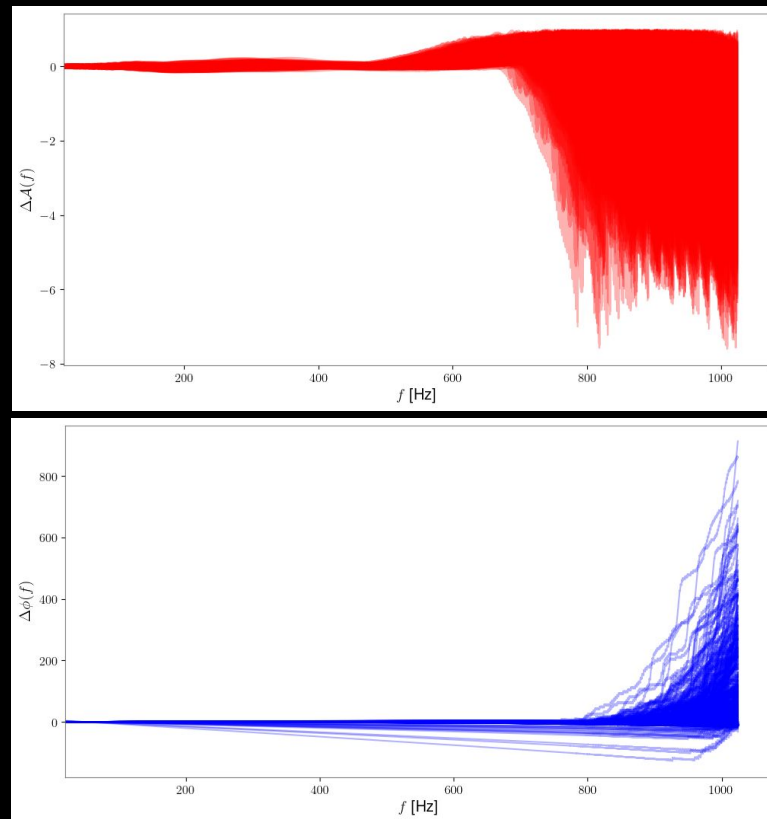
- Waveform models end the waveform at different frequencies
 - Often caused by tidal effects in binary neutron stars
- This causes discontinuities to occur in the amplitude difference
- Cut the waveform differences at the discontinuity

$$\lim_{\mathcal{A}_1(f) \rightarrow 0} \left(\frac{\mathcal{A}_2(f) - \mathcal{A}_1(f)}{\mathcal{A}_1(f)} \right) = \infty$$

$$\lim_{\mathcal{A}_2(f) \rightarrow 0} \left(\frac{\mathcal{A}_2(f) - \mathcal{A}_1(f)}{\mathcal{A}_1(f)} \right) = -1$$

Waveform Model Differences

- Assumes no usable data after the discontinuity
- Each set ends at a different frequency
- Good for looking at the distribution of waveform differences
- Cannot be used for our prior
 - Each parameter must have data



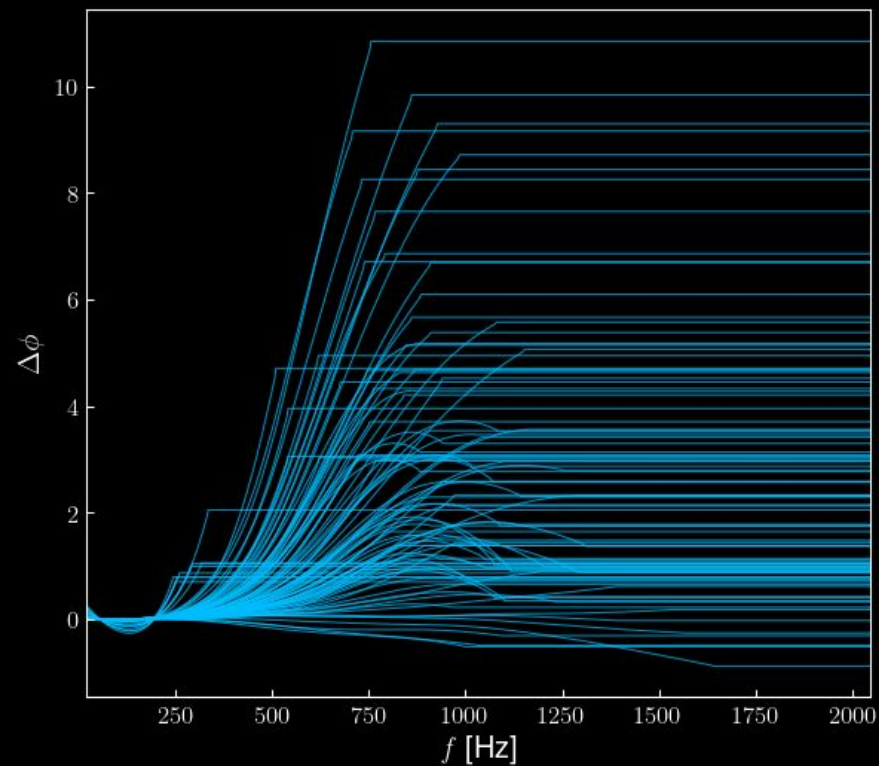
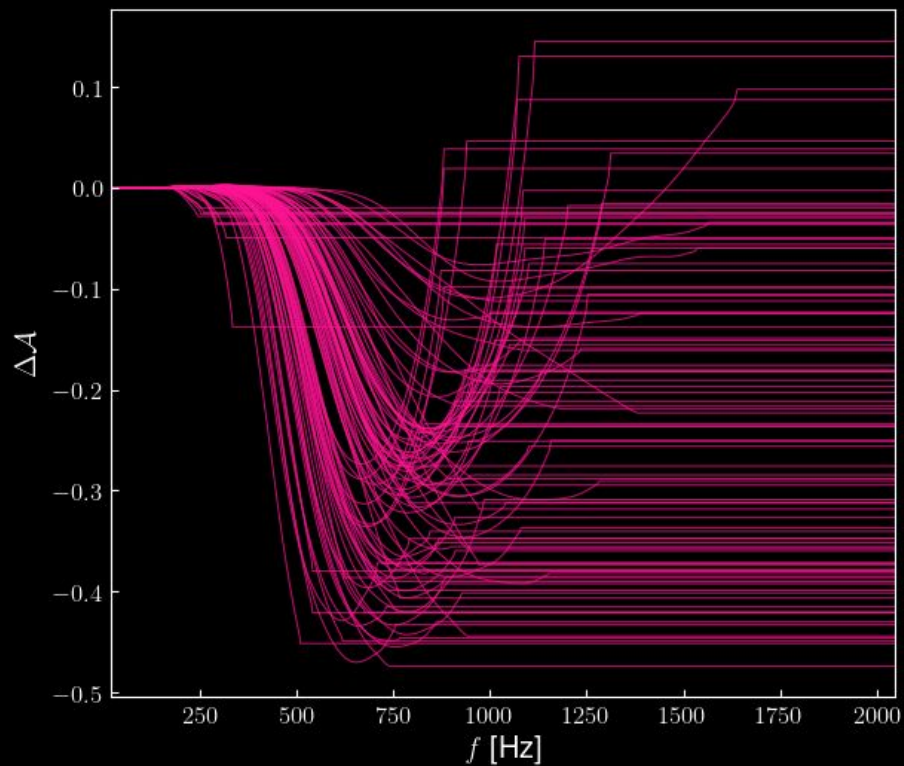
Waveform Model Differences

- We need a fixed range of frequencies
- Every frequency needs data
- Hold the waveform differences constant after the discontinuity
 - No information after the discontinuity, so we cannot do anything else
- We define the waveform model differences

$$\Delta\mathcal{A}_\mu(f; \theta) = \begin{cases} \Delta\mathcal{A}(f; \theta) & f \leq f_{COR} \\ \Delta\mathcal{A}(f_{COR}; \theta) & f > f_{COR} \end{cases}$$

$$\Delta\phi_\mu(f; \theta) = \begin{cases} \Delta\phi(f; \theta) & f \leq f_{COR} \\ \Delta\phi(f_{COR}; \theta) & f > f_{COR} \end{cases}$$

Waveform Model Differences



Defining the Prior

- Define a prior distribution for the waveform differences
- Cannot be too large
 - Model Correction Function, ν , can make any set of parameters match the data if allowed
 - High degeneracy
- Cannot be a uniform prior
 - Constricted to a narrow region of parameter space
 - If expanded, degeneracies become likely
- We choose a Gaussian prior
 - Sampler can reach any waveform difference
 - Decreased probability away from the mean lowers the chances of finding degeneracies
- Define the means and standard deviations

Defining the Prior

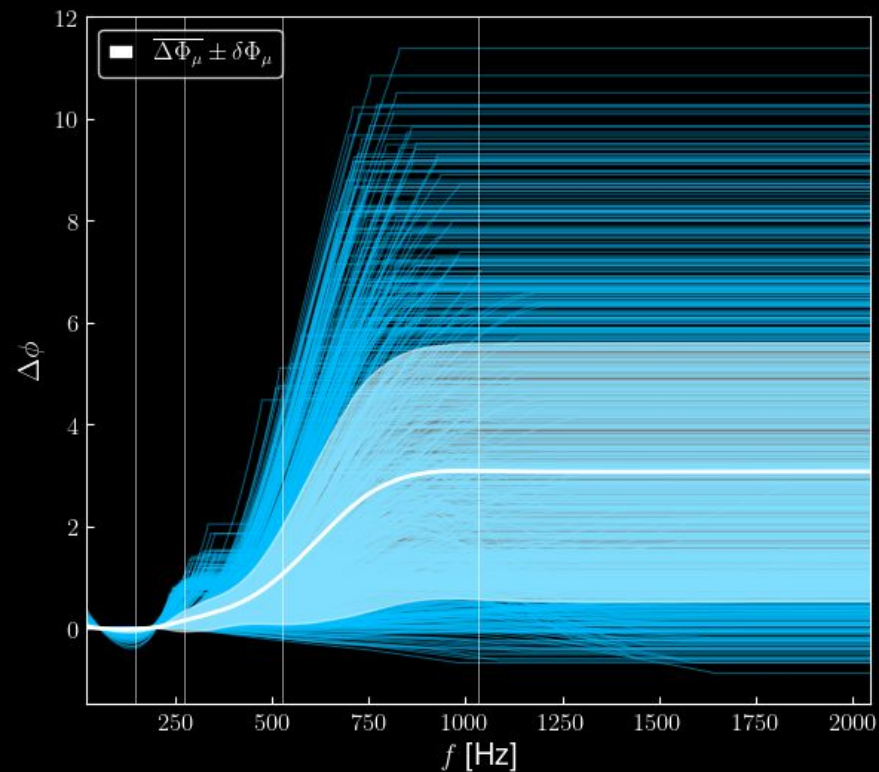
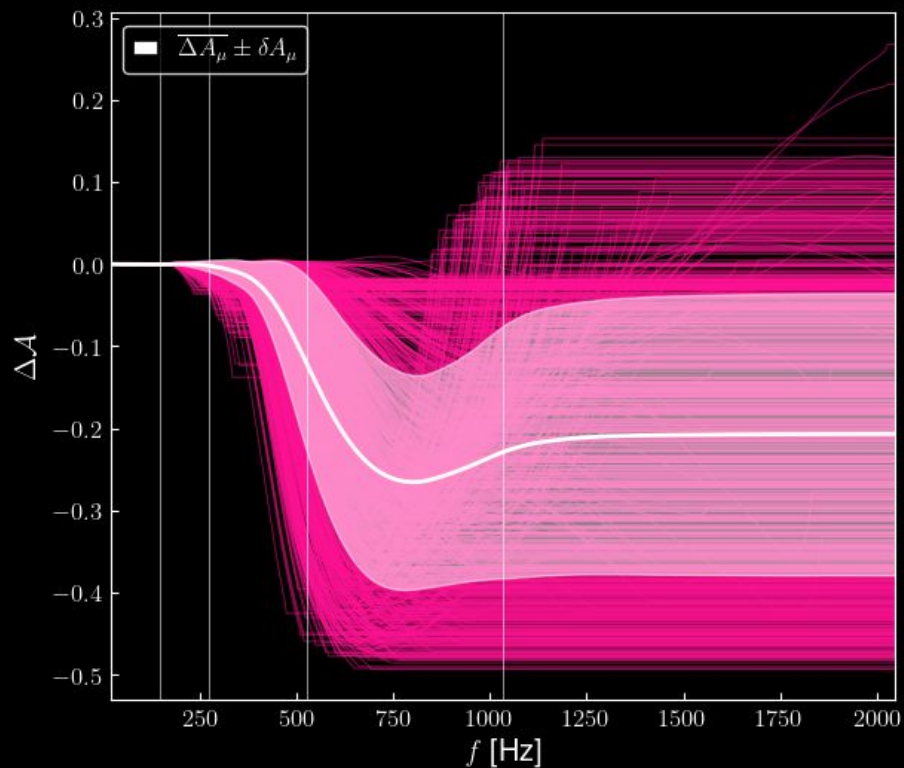
$$\overline{\Delta\mathcal{A}_\mu}(f) = \frac{\sum_{i=1}^N (\Delta\mathcal{A}_\mu(f; \theta_i))}{N}$$

$$\delta\mathcal{A}_\mu(f) = \sqrt{\frac{\sum_{i=1}^N (\Delta\mathcal{A}_\mu(f; \theta_i) - \overline{\Delta\mathcal{A}_\mu}(f))^2}{N}}$$

$$\overline{\Delta\phi_\mu}(f) = \frac{\sum_{i=1}^N (\Delta\phi_\mu(f; \theta_i))}{N}$$

$$\delta\phi_\mu(f) = \sqrt{\frac{\sum_{i=1}^N (\Delta\phi_\mu(f; \theta_i) - \overline{\Delta\phi_\mu}(f))^2}{N}}$$

Defining the Prior

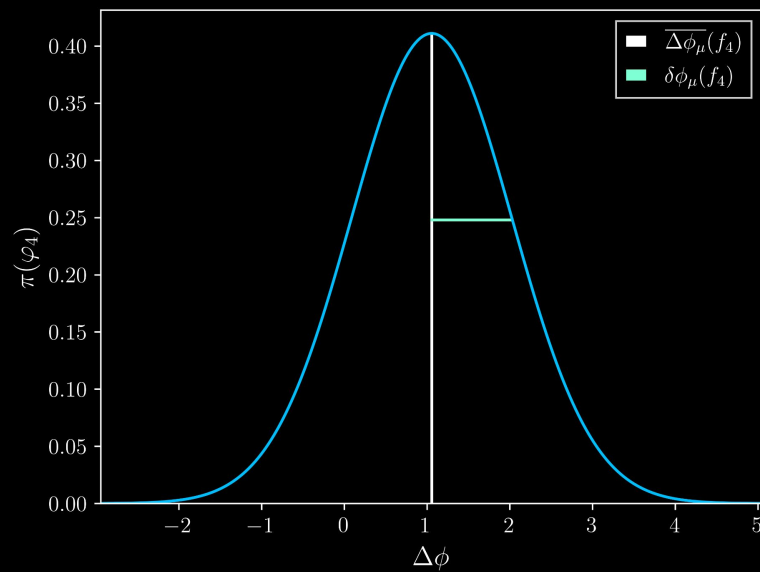
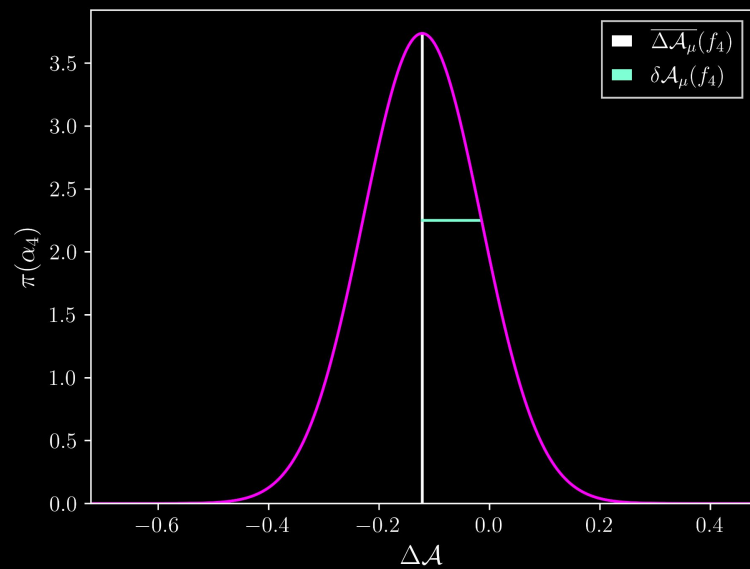


Defining the Prior

$$\pi(\alpha_k) = \frac{(2\pi)^{-\frac{1}{2}}}{\delta \mathcal{A}_\mu(f_k)} \exp \left[-\frac{1}{2} \left(\frac{\Delta \mathcal{A} - \overline{\Delta \mathcal{A}_\mu}(f_k)}{\delta \mathcal{A}_\mu(f_k)} \right)^2 \right]$$

$$\pi(\varphi_k) = \frac{(2\pi)^{-\frac{1}{2}}}{\delta \phi_\mu(f_k)} \exp \left[-\frac{1}{2} \left(\frac{\Delta \phi - \overline{\Delta \phi_\mu}(f_k)}{\delta \phi_\mu(f_k)} \right)^2 \right]$$

Defining the Prior



Everything we need to start parameter estimation:

$$\mu_{\mathcal{C}}(f; \theta, \alpha, \varphi) = \mu(f; \theta) \cdot \nu(f; \alpha, \varphi)$$

$$\pi(\alpha_k) = \frac{(2\pi)^{-\frac{1}{2}}}{\delta \mathcal{A}_{\mu}(f_k)} \exp \left[-\frac{1}{2} \left(\frac{\Delta \mathcal{A} - \overline{\Delta \mathcal{A}_{\mu}}(f_k)}{\delta \mathcal{A}_{\mu}(f_k)} \right)^2 \right]$$

$$\pi(\varphi_k) = \frac{(2\pi)^{-\frac{1}{2}}}{\delta \phi_{\mu}(f_k)} \exp \left[-\frac{1}{2} \left(\frac{\Delta \phi - \overline{\Delta \phi_{\mu}}(f_k)}{\delta \phi_{\mu}(f_k)} \right)^2 \right]$$

Preliminary Results

Parameter Estimation Test

- Binary Black Holes (BBH)
- Define a set of prior distributions
- Choose a set of injection parameters
- Generate a waveform to be our data
- Perform two PE runs:
 1. Null-Hypothesis (NHP) Run
 - Assumes no uncertainties
 - Does not perform the correction
 2. Waveform Uncertainty (WFU) Corrected Run
 - Takes uncertainties into account
 - Performs the correction
- Compare the posteriors

Preparation - Source Parameter Priors

$$\pi(\mathcal{M}) = \mathcal{U}(10, 40)$$

$$\pi(q) = \mathcal{U}(0.34, 1.00)$$

$$\pi(\chi_1) = \mathcal{U}(-0.5, 0.5)$$

$$\pi(\chi_2) = \mathcal{U}(-0.5, 0.5)$$

$$\pi(\mathcal{L}_D) = \mathcal{U}(100, 1500)$$

$$\pi(t_c) = \mathcal{U}(-0.2, +0.2)$$

$$\pi(\phi) = \mathcal{U}(0, 2\pi)$$

$$\pi(\phi_{JL}) = \delta(0.4543)$$

$$\pi(\phi_{12}) = \delta(1.2242)$$

$$\pi(\theta_{JN}) = \delta(0.4874)$$

$$\pi(\delta) = \delta(-1.0108)$$

$$\pi(\alpha_r) = \delta(1.7475)$$

$$\pi(\psi) = \delta(2.6597)$$

Preparation - Waveform Uncertainty Priors

- Ignore Amplitude Uncertainty
 - Little effect on the posterior
 - Saves computation time

$$\pi(\alpha_1) = \delta(0.0)$$

$$\pi(\alpha_2) = \delta(0.0)$$

$$\pi(\alpha_3) = \delta(0.0)$$

$$\pi(\alpha_4) = \delta(0.0)$$

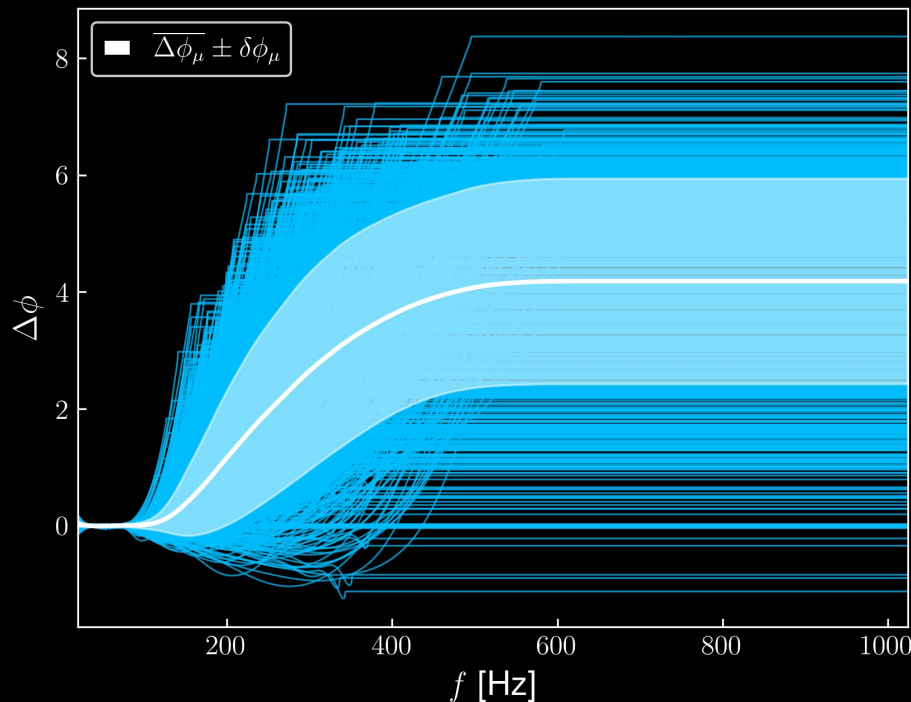
$$\pi(\alpha_5) = \delta(0.0)$$

$$\pi(\alpha_6) = \delta(0.0)$$

Preparation - Waveform Uncertainty Priors

$$\tilde{h}_1(f) = \mu_{IMR}(f; \theta)$$

$$\tilde{h}_2(f) = \mu_{EOB}(f; \theta)$$



Preparation - Waveform Uncertainty Priors

$$\pi(\varphi_1) = \delta(0.0)$$

$$\pi(\varphi_2) = \delta(0.0)$$

$$\pi(\varphi_3) = \mathcal{N}(0, 0.4743)$$

$$\pi(\varphi_4) = \mathcal{N}(0, 1.5912)$$

$$\pi(\varphi_5) = \mathcal{N}(0, 1.7930)$$

$$\pi(\varphi_6) = \mathcal{N}(0, 1.7568)$$

Preparation

- Choose an injection, or a fixed model to act as our data
- These are our true parameter values
- Correction posterior will ideally return these

$$\theta^{\text{inj}} = \left\{ \begin{array}{l} \mathcal{M} = 21.0451 \text{ M}_{\odot} \\ q = 0.9740 \\ \chi_1 = 0.3802 \\ \chi_2 = -0.2089 \\ \mathcal{L}_D = 1295.781 \text{ Mpc} \\ t_c = -0.0772 \text{ s} \\ \phi = 3.9448 \\ \phi_{JL} = 0.4543 \\ \phi_{12} = 1.2242 \\ \theta_{JN} = 0.4874 \\ \delta = -1.0108 \\ \alpha_r = 1.7475 \\ \psi = 2.6597 \end{array} \right.$$

$$\alpha^{\text{inj}} = \left\{ \begin{array}{l} \alpha_1 = 0.0 \\ \alpha_2 = 0.0 \\ \alpha_3 = 0.0 \\ \alpha_4 = 0.0 \\ \alpha_5 = 0.0 \\ \alpha_6 = 0.0 \end{array} \right.$$
$$\varphi^{\text{inj}} = \left\{ \begin{array}{l} \varphi_1 = 0.0156 \\ \varphi_2 = 0.0429 \\ \varphi_3 = 0.0626 \\ \varphi_4 = 1.3377 \\ \varphi_5 = -1.2659 \\ \varphi_6 = 1.7418 \end{array} \right.$$

Null-Hypothesis Run

- Assumes that there are no uncertainties
 - Standard waveform model and likelihood
- Posterior distributions will be pushed away from the injected values by the waveform differences

$$\mathcal{L}_{\emptyset}(h^{\text{inj}}|\theta) = \prod_j \frac{1}{2\pi S_n(f_j)} \exp \left(-2\Delta f \frac{|\tilde{h}^{\text{inj}}(f_j) - \mu(f_j; \theta)|^2}{S_n(f_j)} \right)$$

Waveform Uncertainty-Corrected Run

- Samples the waveform uncertainty parameters
- Matching the waveform differences will give an improved waveform model
- Ideally, this returns the exact injection parameters

$$\mathcal{L}_C(h^{\text{inj}}|\theta, \alpha, \varphi) = \prod_j \frac{1}{2\pi S_n(f_j)} \exp \left(-2\Delta f \frac{|\tilde{h}^{\text{inj}}(f_j) - \mu(f; \theta) \cdot \nu(f; \alpha, \varphi)|^2}{S_n(f_j)} \right)$$

Comparing the Posteriors

- Choose the maximum likelihood parameters as the recovered parameters
- Compare these with the injected parameters

$$\mathcal{L}_c(h^{\text{inj}}|\theta_c^{\text{ml}}, \alpha^{\text{ml}}, \varphi^{\text{ml}}) = \max[\mathcal{L}_c(h^{\text{inj}}|\theta, \alpha, \varphi)]$$

$$\mathcal{L}_\emptyset(h^{\text{inj}}|\theta_\emptyset^{\text{ml}}) = \max[\mathcal{L}_\emptyset(h^{\text{inj}}|\theta)]$$

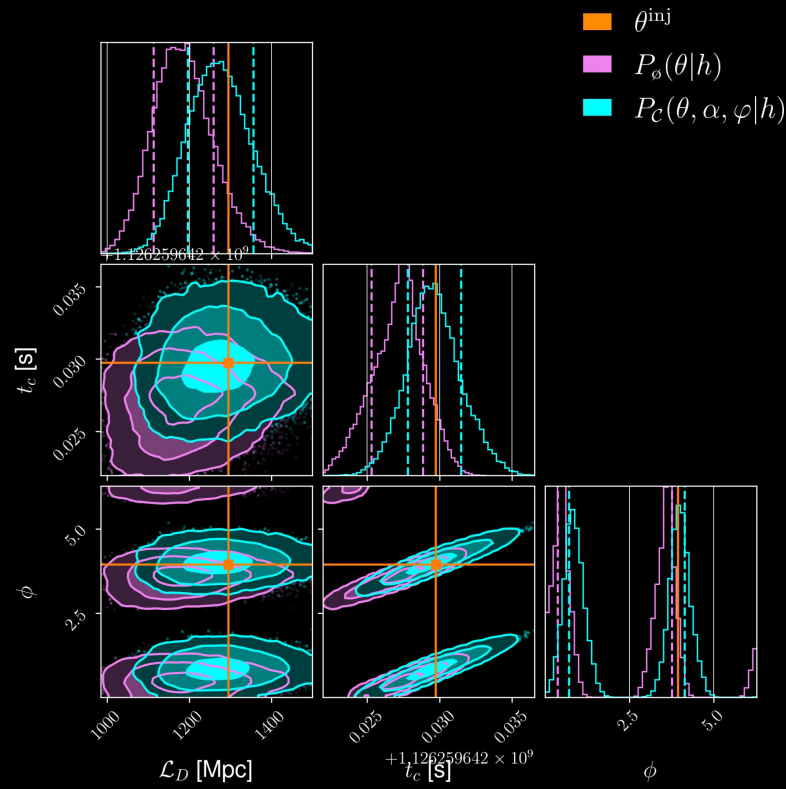
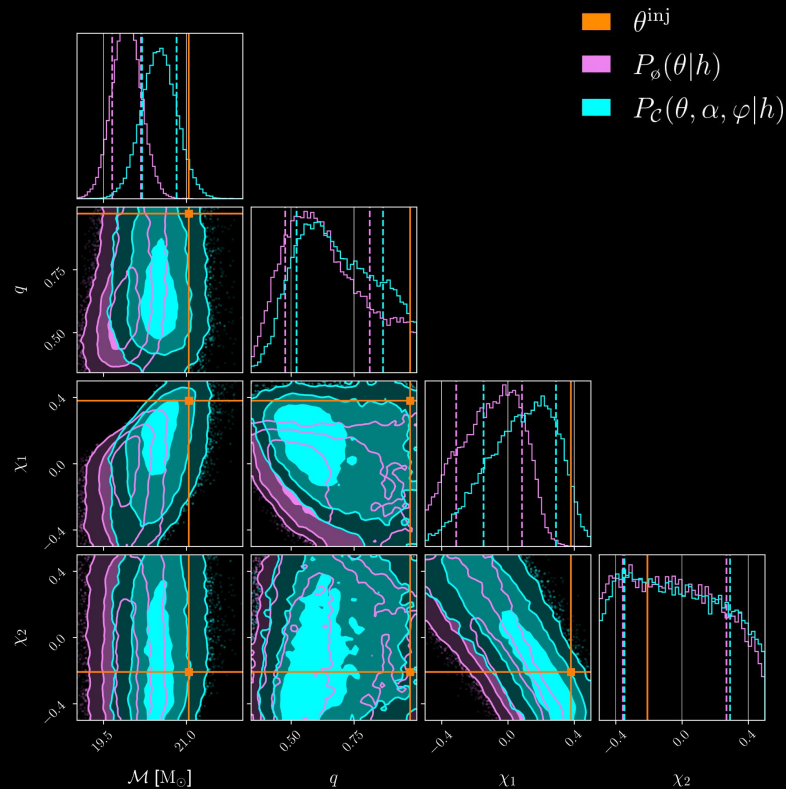
Comparing the Posteriors

- Define Q as a quantitative measure of the effectiveness of the correction
- Compares the distances between the recovered parameters and the the injected parameters

$$Q = 1 - \sqrt{\frac{\sum_p (\theta_c^{\text{ml}}[p] - \theta^{\text{inj}}[p])^2}{\sum_{p'} (\theta_{\emptyset}^{\text{ml}}[p'] - \theta^{\text{inj}}[p'])^2}}$$

$Q=1$	Ideal case; perfect correction
$Q>0$	Improvement over NHP
$Q<0$	Worse than NHP

Comparing the Posteriors



Comparing the Posteriors

